

Recap Lecture 12

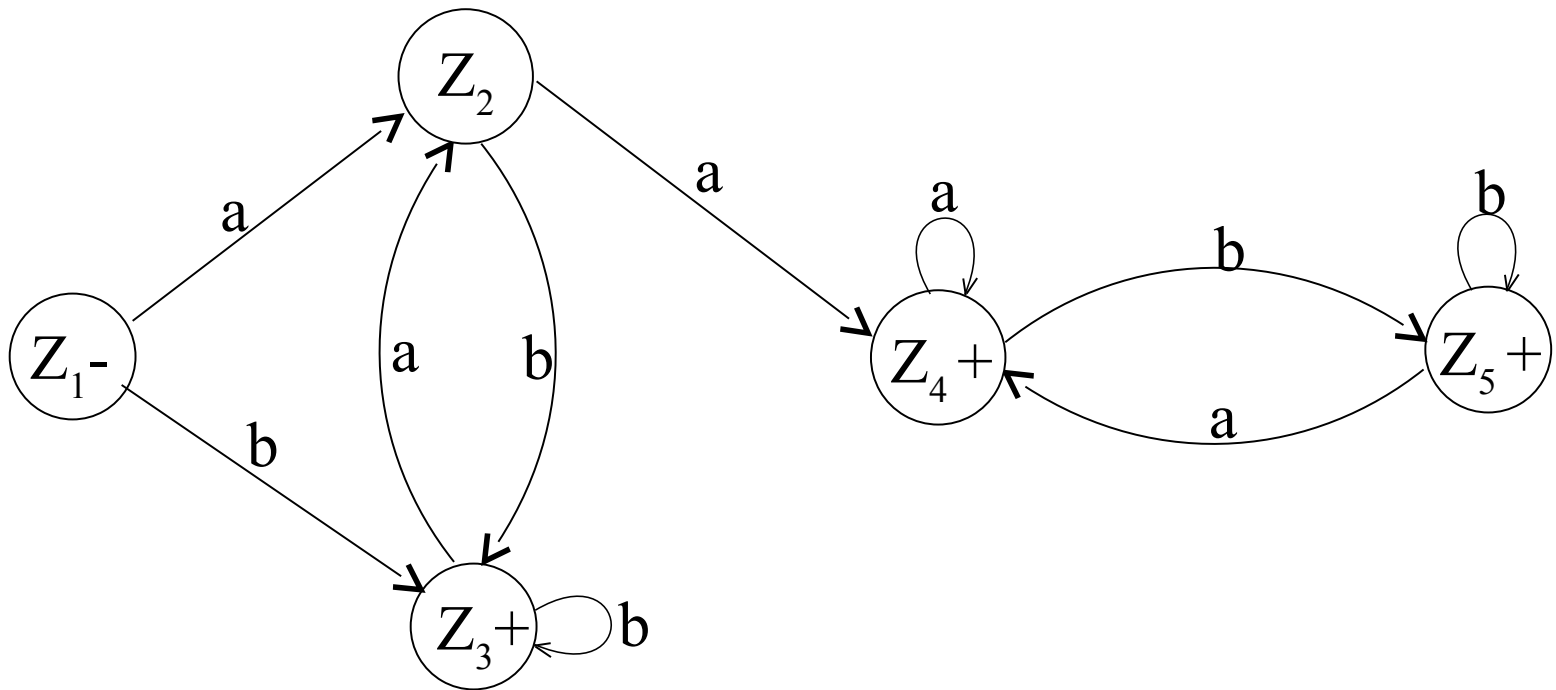


- Examples of writing REs to the corresponding TGs, RE corresponding to TG accepting EVEN-EVEN language, Kleene's theorem part III (method 1: union of FAs), examples of FAs corresponding to simple REs, example of Kleene's theorem part III (method 1) continued

Note

- It may be noted that in the previous example FA_1 contains two states while FA_2 contains three states. Hence the total number of possible combinations of states of FA_1 and FA_2 , in sequence, will be six. For each combination the transitions for both a and b can be determined, but using the method in the example, number of states of FA_3 was reduced to five.

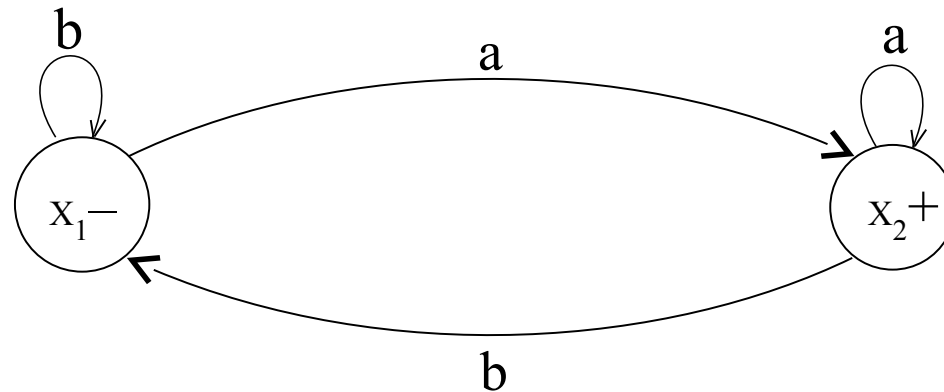
Task



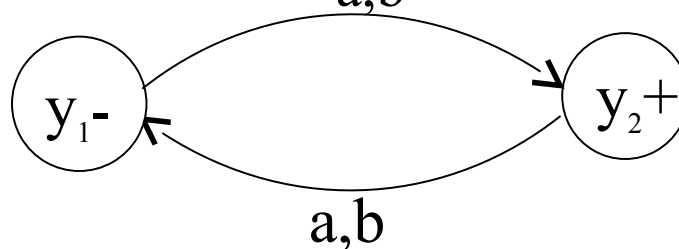
Build an FA equivalent to the previous FA

Example

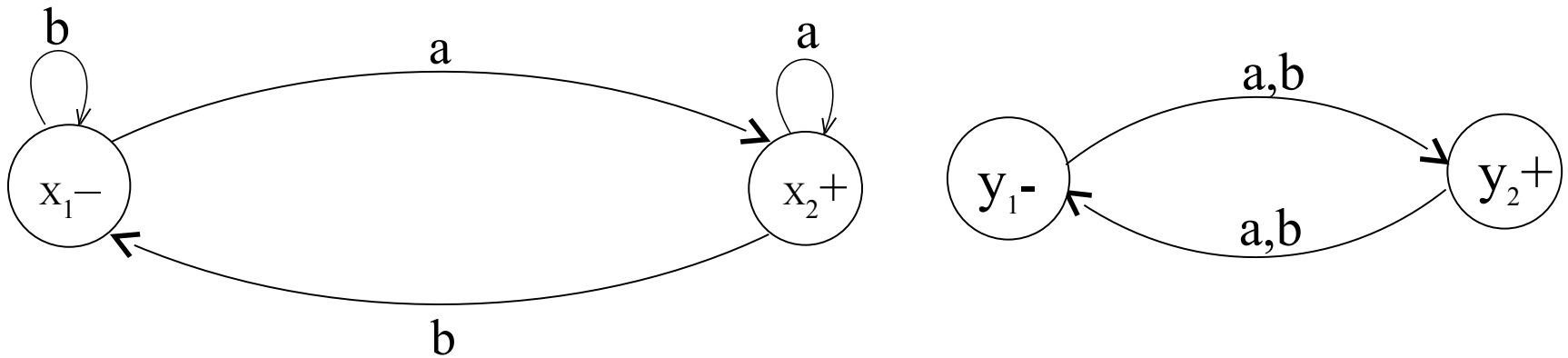
Let $r_1 = (a+b)^*a$ and the corresponding FA_1 be



also $r_2 = (a+b)((a+b)(a+b))^*$ or $((a+b)(a+b))^*(a+b)$ and FA_2 be



Example continued ...

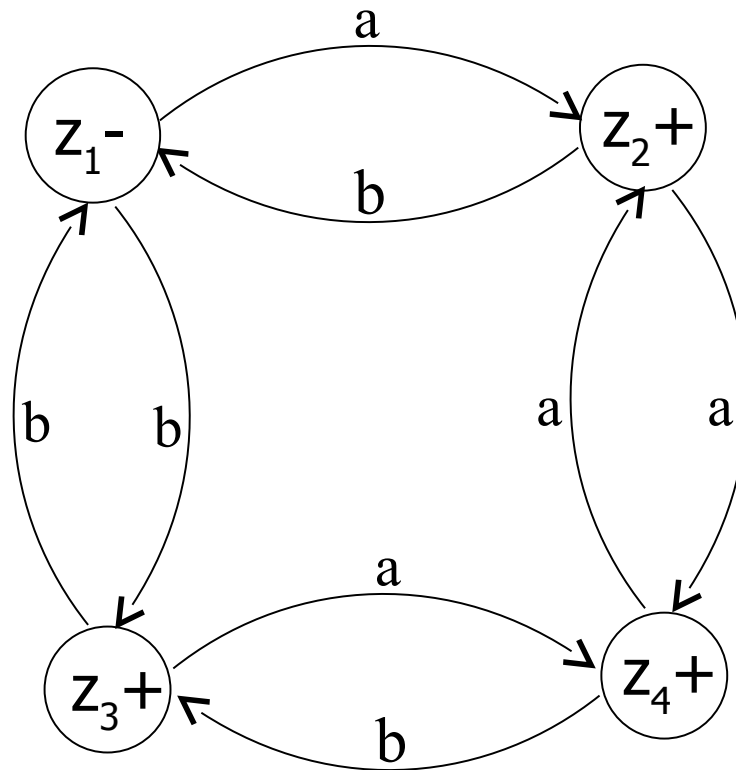


Old States	New States after reading	
	a	b
$z_1 - (x_1, y_1)$	$(x_2, y_2) \sqcup z_2$	$(x_1, y_2) \sqcup z_3$

Example continued ...

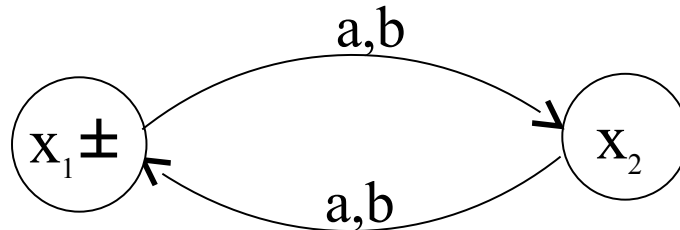
Old States	New States after reading	
	a	b
$z_2 + (x_2, y_2)$	$(x_2, y_1) \rightarrow z_4$	$(x_1, y_1) \rightarrow z_1$
$z_3 + (x_1, y_2)$	$(x_2, y_1) \rightarrow z_4$	$(x_1, y_1) \rightarrow z_1$
$z_4 + (x_2, y_1)$	$(x_2, y_2) \rightarrow z_2$	$(x_1, y_2) \rightarrow z_3$

Example continued ...

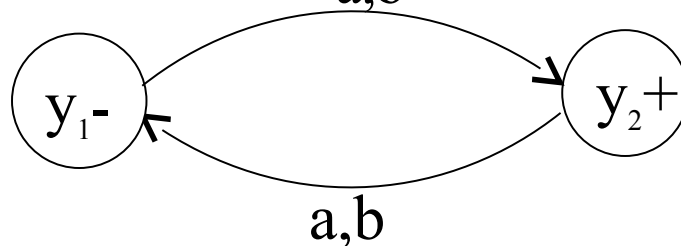


Example

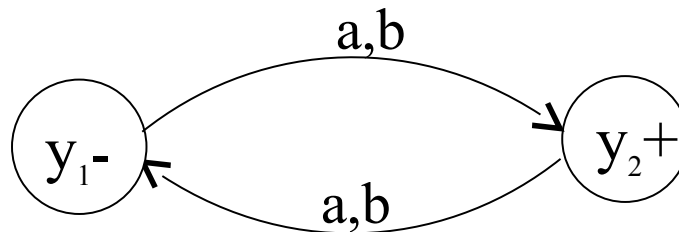
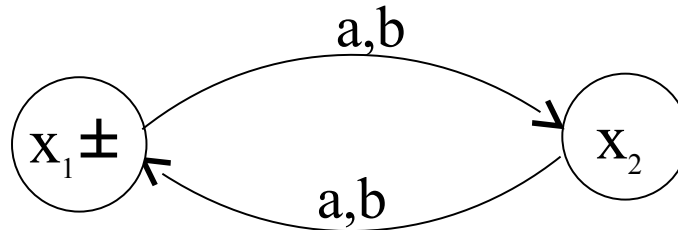
Let $r_1 = ((a+b)(a+b))^*$ and the corresponding FA_1 be



also $r_2 = (a+b)((a+b)(a+b))^*$ or $((a+b)(a+b))^*(a+b)$ and FA_2 be



Example continued ...

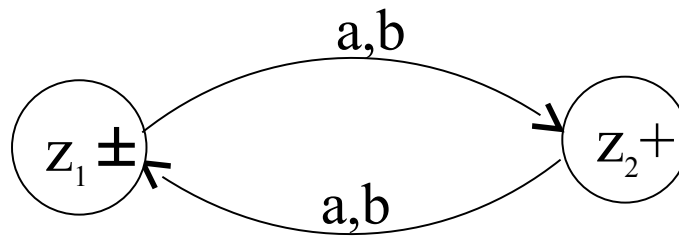


Old States	New States after reading	
	a	b
$z_1 \pm \sqcap (x_1, y_1)$	$(x_2, y_2) \sqcap z_2$	$(x_2, y_2) \sqcap z_2$

Example continued ...

Old States	New States after reading	
	a	b
$z_2 + \square(x_2, y_2)$	$(x_1, y_1) \square z_1$	$(x_1, y_1) \square z_1$

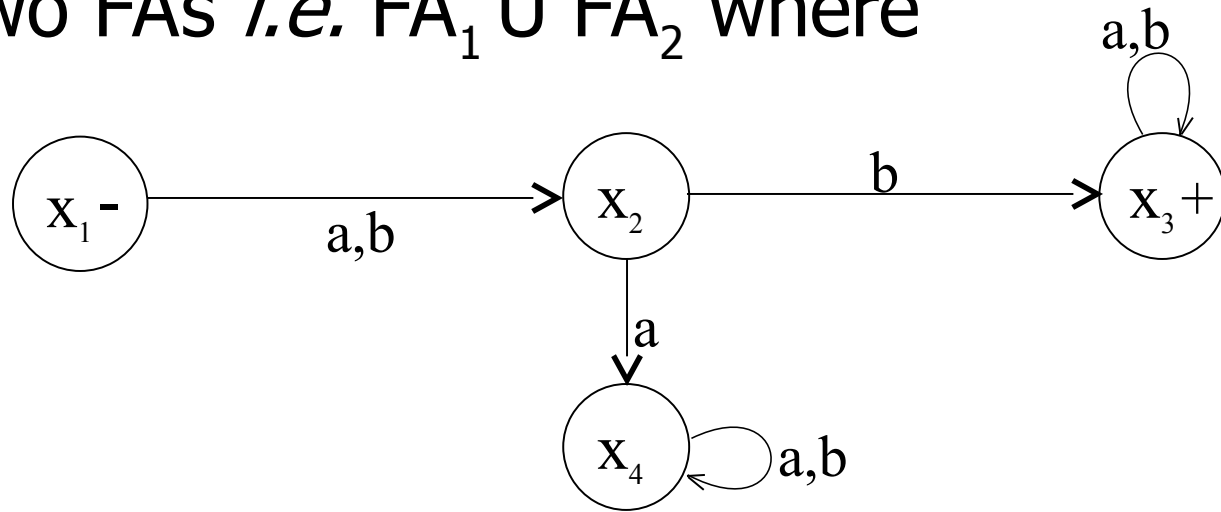
Example continued ...



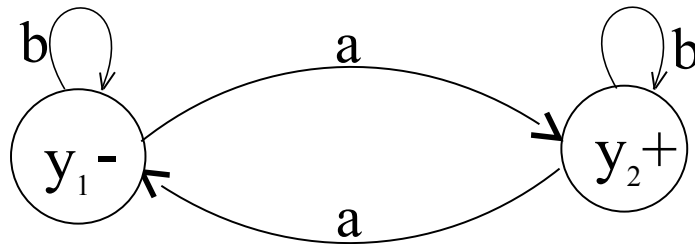
Task

Build an FA corresponding to the union of these two FAs *i.e.* $FA_1 \cup FA_2$ where

FA_1



FA_2



Kleene's Theorem Part III

Continued ...



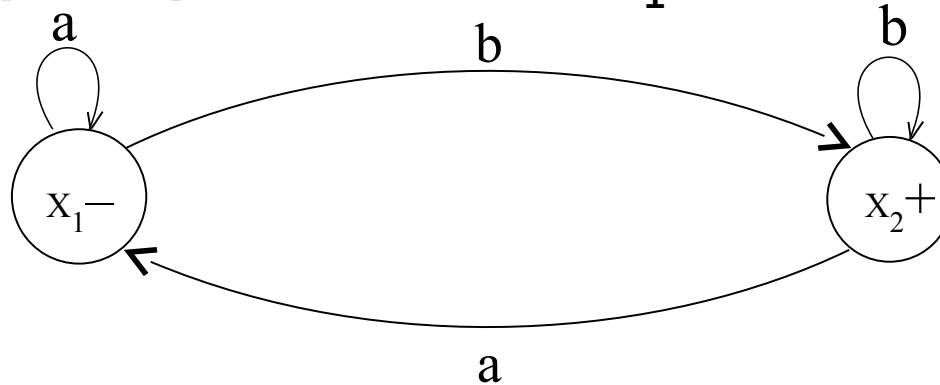
□ Method2 (Concatenation of two FAs):

Using the FAs corresponding to r_1 and r_2 , an FA can be built, corresponding to $r_1 r_2$.

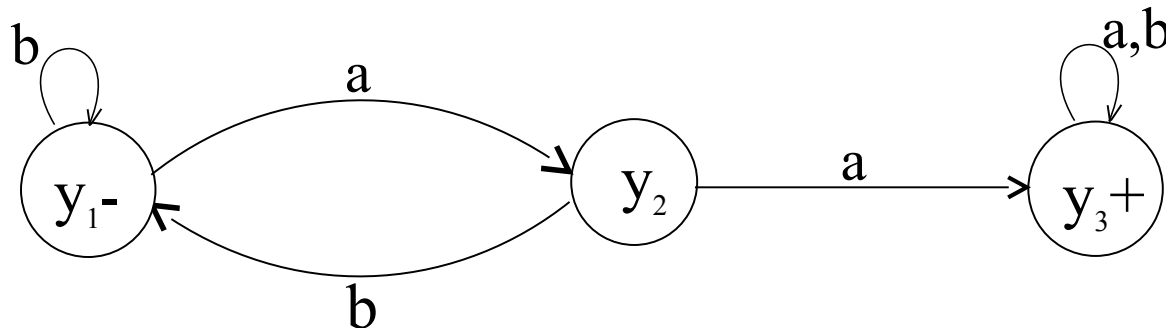
This method can be developed considering the following examples

Example

Let $r_1 = (a+b)^*b$ defines L_1 and FA_1 be



and $r_2 = (a+b)^*aa(a+b)^*$ defines L_2 and FA_2 be



Concatenation of two FAs

Continued ...



Let FA_3 be an FA corresponding to r_1r_2 , then the initial state of FA_3 must correspond to the initial state of FA_1 and the final state of FA_3 must correspond to the final state of FA_2 . Since the language corresponding to r_1r_2 is the concatenation of corresponding languages L_1 and L_2 , consists of the strings obtained, concatenating the strings of L_1 to those of L_2 , therefore ***the moment a final state of first FA is entered, the possibility of the initial state of second FA will be included as well.***

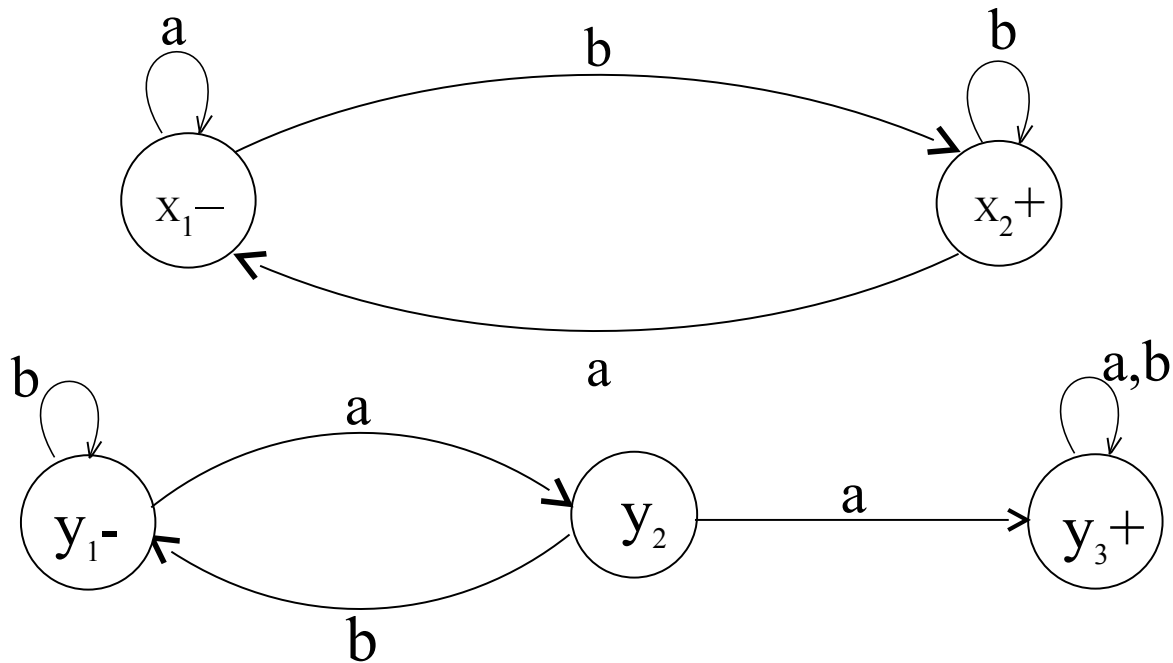
Concatenation of two FAs

Continued ...



Since, in general, FA_3 will be different from both FA_1 and FA_2 , so the labels of the states of FA_3 may be supposed to be $z_1, z_2, z_3, \dots$, where z_1 stands for the initial state. Since z_1 corresponds to the states x_1 , so there will be two transitions separately for each letter read at z_1 . It will give two possibilities of states which correspond to either z_1 or different from z_1 . This process may be expressed in the following transition table for all possible states of FA_3

Example continued ...

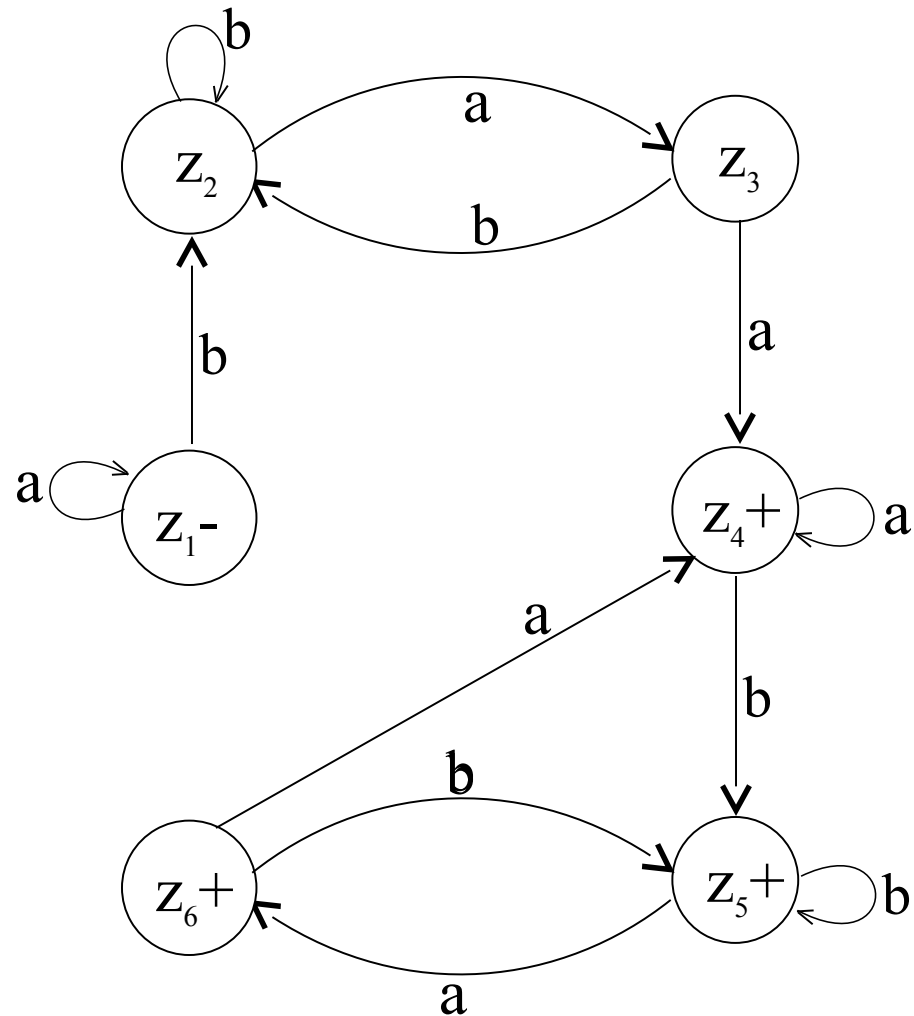


Old States	New States after reading	
	a	b
$z_1^- \sqcap x_1$	$x_1 \sqcap z_1$	$(x_2, y_1) \sqcap z_2$

Example continued ...

Old States	New States after reading	
	a	b
$z_2 \sqcap (x_2, y_1)$	$(x_1, y_2) \sqcap z_3$	$(x_2, y_1) \sqcap z_2$
$z_3 \sqcap (x_1, y_2)$	$(x_1, y_3) \sqcap z_4$	$(x_2, y_1) \sqcap z_2$
$z_4^+ \sqcap (x_1, y_3)$	$(x_1, y_3) \sqcap z_4$	$(x_2, y_1, y_3) \sqcap z_5$
$z_5^+ \sqcap (x_2, y_1, y_3)$	$(x_1, y_2, y_3) \sqcap z_6$	$(x_2, y_1, y_3) \sqcap z_5$
$z_6^+ \sqcap (x_1, y_2, y_3)$	$(x_1, y_3) \sqcap z_4$	$(x_2, y_1, y_3) \sqcap z_5$

Example continued ...



Summing Up



- Examples of Kleene's theorem part III
(method 1) continued ,Kleene's theorem part III
(method 2: Concatenation of FAs),
- Example of Kleene's theorem part III
(method 2 : Concatenation of FAs)